SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Project Report on

"FABRICATION OF ADJUSTABLE PILLAR CLIMBER"

submitted in partial fulfillment of the requirement for the completion of V semester of

BACHELOR OF ENGINEERING

in

MECHANICAL ENGINEERING

Submitted by

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2025-26

SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CERTIFICATE

Certified that the mini project work entitled "FABRICATION OF ADJUSTABLE PILLAR CLIMBER" is a bonafide work carried out by Satvik J Sheelavant(1SI23ME083), Siddarth S (1SI23ME093), Veeresh S G (1SI23ME108) and C Yashwanth (1SI24ME402) in partial fulfillment for the completion of VI Semester of Bachelor of Engineering in Mechanical Engineering from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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Course Outcomes

CO1: : Identify a relevant engineering problem through a systematic literature review and by understanding current advancements in mechanical and design engineering.

CO2: To consolidate the literature search to identify issues/gaps and formulate the engineering problem

CO3: To prepare project schedule for the identified design methodology and engage in budget analysis, and share responsibility for every member in the team

CO4: To provide sustainable engineering solution considering health, safety, legal, cultural issues and also demonstrate concern for environment

CO5: To identify and apply the mathematical concepts, science concepts, engineering and management concepts necessary to implement the identified engineering problem

CO6: To select the engineering tools/components required to implement the proposed solution for the identified engineering problem

CO7: To analyze, design, and implement optimal design solution, interpret results of experiments and draw valid conclusion

CO8: To demonstrate effective written communication through the project report, the one-page poster presentation, and preparation of the video about the project and the four page IEEE/Springer/ paper format of the work

CO9: To engage in effective oral communication through power point presentation and demonstration of the project work

CO10:To demonstrate compliance to the prescribed standards/ safety norms and abide by the norms of professional ethics

CO11: To perform in the team, contribute to the team and mentor/lead the team

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

POs: PO1: Engineering Knowledge, PO2: Problem analysis, PO3: Design/Development of solutions, PO4: Conduct investigations of complex problems, PO5: Modern tool usage, PO6: Engineer and the world, PO7: Ethics, PO8: Individual and collaborative team work, PO9: Communication, PO10: Project management and finance, PO11: Lifelong learning

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1											3		
CO-2		3											
CO-3											3		
CO-4						3	3						
CO-5	3	3											
CO-6					3								
CO-7			3	3									
CO-8										3			
CO-9										3			
CO-10								3					
CO-11									3				

Abstract

An abstract is a concise the increasing need for safe and reliable access to vertical structures such as building pillars, electric poles, has created a demand for climbing systems that are efficient, safe, and cost-effective. Traditional methods such as ladders are often unstable, require manual support, and pose a high risk of falls during maintenance operations. Recognizing these limitations, the project focuses on developing a manually operated adjustable pillar climber that provides a safer alternative for inspection, servicing, and The primary objective of this project is to model and fabricate a lightweight, adjustable climbing mechanism capable of securely gripping surfaces while supporting a user's weight during vertical movement. The project aims to achieve simplicity, affordability, and operational safety while incorporating ergonomic considerations and material strength requirements..

The implementation involved conceptual model, model development, analysis, and fabrication. SolidWorks was used to create the 3D design and verify basic dimensions and assembly feasibility. The prototype was fabricated using materials such as stainless steel, and rubber pads for grip, while standard tools like welding equipment, grinders, and drilling machines were used during construction. The final model operates through a manual gripping system that enables controlled and safe upward and downward movement on vertical pillars. The fabricated system demonstrates a viable approach for safe manual pillar climbing, and serves as a foundation for future improvements such as motorized operation or remote control functionality.

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Chapter 1

Introduction

1.1 Motivation

The fabrication of an adjustable pillar climber addresses an essential need in modern construction, maintenance, and inspection environments where vertical structures such as pillars, poles, and columns must be accessed safely and efficiently. Traditional methods, including fixed ladders, scaffolding, or manual climbing, often pose risks, require time-consuming setup, and lack adaptability to pillars of varying lengths and surface conditions. An adjustable pillar climber provides a practical solution by enabling secure, ergonomic, and rapid ascent and descent on different types of vertical supports.With increasing demand for maintenance of streetlights, industrial chimneys, and utility installations, there is a growing emphasis on worker safety and operational convenience. An adjustable climber not only enhances accessibility but also minimizes the risk of falls and injuries by integrating stable gripping mechanisms and adjustable features. As industries continue to adopt compact, user-friendly, and cost-effective climbing tools, developing a reliable adjustable pillar climber becomes vital for improving overall efficiency and reducing downtime during routine inspections and repairs.

1.2 Problem Identification and Requirement Study

Manual pillar climbing for maintenance and inspection work involves high risk of slipping, falling, and physical strain. Skilled labor is required, and safety equipment is often insufficient. Hence, there is a need for a simple, cost-effective, and manually operated pillar climbing device that can reduce human effort and improve safety during climbing operations.

1.3 Organisation of the report

This report is organised into the following chapters for systematic presentation and clarity:

Chapter 1: Introduction

This invention discusses the background, motivations, problem description, objectives, and scope, which identify that there was a necessity for developing an adjustable pillar climber.

Chapter 2: Literature Review

Discusses current climbing mechanism designs, pillar climbing devices, adjustable link systems, and designs with a focus on safety. Points out some of the gaps in research that are filled in this project.

Chapter 3: Materials and Methodology

Describes how to select a frame, a grip system, and an adjustment mechanism. Fabrication of mechanism, mechanism types, and flow.

Chapter 4: Experimental Setup

The equipment used to test stability, load capacity, adjustability, and grip performance, along with what these tests aim to accomplish.

Chapter 5: Results and Discussion

Presents and discusses performance results, with a comparison of design iterations and adjustment parameters for different loading conditions.

Chapter 6: Conclusion

Points out findings, emphasizing enhancements in safety, efficiency, and functionality of usage for the adjustable pillar climber.

Chapter 7: Future Scope

Proposes recommendations that relate to motorized climbing, enhanced ergonomics, automation, improved grip materials, and design optimization.

Chapter 2

Literature Survey

2.1 Introduction

With the rising demand for safe, efficient, and adaptable climbing devices, there is a need for developing adjustable pillar climbers used in various industrial and service applications. Workers generally adopt manual methods of climbing vertical pillars in different professions, such as electrical maintenance, construction, telecom installation, cleaning services, and industrial inspection. These traditional methods of climbing usually involve ladders, ropes, harnesses, or belt-based climbing systems. While these methods are widely in practice, they all involve high levels of physical strain, fatigue, and accidental risks to workers. In this regard, available literature justifies the necessity for a device on adjustable pillar climbing with improved safety, stability, and usability for operators working at heights.

Research on structural climbing devices focuses on making a mechanism that could adapt to different diameters of pillars without losing grip strength or comfort for a user. Climbing devices, inspired by both natural climbing organisms and industrial gripping systems, have identified adaptability as one aspect that boosts climbing efficiency. Research in the field of mechanical gripping mechanisms demonstrates that multi-contact gripping, passive compliance, and construction of flexible frames ensure stability on varying shapes of a pillar. The devices with hinged arms, flexible tension locks, and geometrically optimized arms reported an increase in adaptability and decreased operating loads on

2.2 Kalra and Kumar Jain (2019) [1]

The journal discusses the fabrication of an adjustable pillar climbing mechanism for performance evaluation with a view to enhancing vertical maintenance operations. It contains adjustable semi-circular jaws, a mechanical screw-based clamping system, and friction-enhancing contact surfaces. Optimizations of clamping forces, material selection, and mechanical stability have been emphasized by the authors in great detail, which are essentially related to safe climbing on smooth cylindrical surfaces. Mild steel and aluminum

were chosen for this purpose because of their excellent strength-to-weight ratio, good machinability, and capability of withstanding structural integrity under variable loads.

The paper also includes detailed calculations involving frictional force, normal reaction forces, and the required torque for screw tightening. Tests conducted on prototype models have shown that rubber pads increase reliability in grip by a significant margin through increased friction coefficients. The study concluded that mechanical climbers offer an alternative low-cost means that is safe to compare to automated robotic systems, more so in developing regions or those with low infrastructure. The paper is relevant to our project because it possesses design principles similar to those our project employs, as well as structural requirements.

2.3 Dileep et al. (2011) [2]

This research encompasses the design and development of a mechanical tree-climbing device, which, although for coconut trees, uses the same principle applied to adjustable pillar climbers. The system uses alternating foot-hold and hand-hold clamps with rubber padding to create frictional grip. The authors have highlighted the problems workers experience in climbing long vertical surfaces: fatigue, safety concerns, and improper gripping due to smooth tree bark or dust.

The mechanism works by the principle of alternating load transfer between the clamps. The paper includes the relevant mechanical calculations of friction, clamping pressure, and the influence of users' weight on stability. Fabrication was done using mild steel and aluminum sheets with welded joints and bolted fasteners. Rubber pads served to improve grip consistency. The findings of this work support our approach regarding frictional grip and safety factors pertinent to our project on adjustable pillar climbing.

2.4 Harikrishna et al. (2013) [3]

This paper presents a robot-based mechanism for climbing specifically on tree trunks. While robotic in nature, the gripping mechanism and consideration of loads, along with friction-based design, show a strong affinity to our concept of an adjustable pillar climber. Compliant jaws are used in the system, which self-adjust tThis chapter includes summary of the findings/surveys that are carried as a ground work for the project. Findings/survey are to be from peer reviewed journals/conferences such as IEEE, IET, etc. .

This illustrates how to cite a reference. This is the first [1] reference cited. To site a

second [2] reference, it's very easy. Refer bibliography.tex file to know , how to cite the reference.

2.5 Mani Jothilingam (2014) [4]

This journal introduces COCOBOT, a coconut-tree climbing robot designed for safe harvesting and trunk scaling operations. The gripping method includes a circular clamp with adjustable leaf-like extensions, enabling adaptation to various tree diameters. The authors analyze different climbing geometries and mathematically derive formulas for required grip force, load distribution, and safety margins.

Fabrication involved laser-cut aluminum, welded steel supports, and composite-material gripping pads. Multiple tests were performed to evaluate slippage, frictional stability, and load capacity. This work supports our project by offering insights into optimum jaw design, the impact of surface texture on frictional behavior, and the structural balance between weight and performance. The paper reinforces the need for redundancy mechanisms in climbing systems.

2.6 Yang et al. (2021) [5]

This study proposes a surrounding-type climbing and pruning robot for tree maintenance. The climbing system fully encloses the trunk using a three-jaw adjustable clamping mechanism. Mechanical simulations and finite element studies were used to optimize the structural frame and gripping force requirements. The authors highlight the relationship between clamping symmetry, load distribution, and climbing stability.

The prototype utilized aluminum frame segments, nitrile rubber pads, and steel reinforcement components. Testing included evaluating climbing speed, slip resistance, and weight-bearing capacity. Although motorized, the detailed analysis of clamping geometry, jaw adaptability, and frictional behavior provides significant theoretical support for our adjustable pillar climber. This journal also emphasizes the importance of safety features and fail-proof gripping mechanisms.

Chapter 3

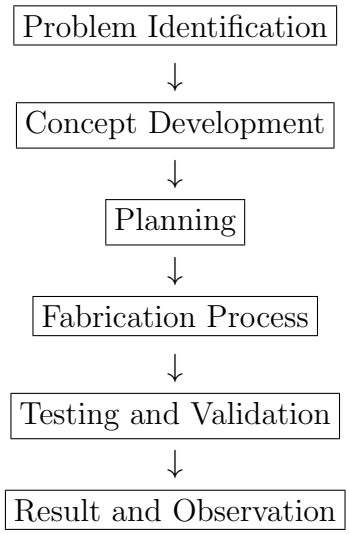
Objective and Methodology

3.1 Objective of the project

- To fabricate the Adjustable climbing model using lightweight and durable materials like aluminum or mild steel or stainless steel.

3.2 Methodology

Figure 3.1: Methodology Flow Chart



3.3 CAD Modelling and Design Finalization

Using CAD software, each component was modeled in 3D:

- All parts of the pillar climber were drawn using CAD software before fabrication.
- Main parts like frame, clamp, screw, and belt support were designed separately.
- Dimensions were chosen to suit different pillar sizes and easy use.
- The complete assembly was checked to ensure proper fit and movement.

Design Analysis Included:

- The design was checked to safely carry the user's body weight.
- Gripping force of the clamp was reviewed to avoid slipping on the pillar.
- Load on the frame and screw during climbing was considered.
- Stability of the climber was verified for safe upward and downward movement.

3.4 Material and Component Selection

Selection was based on strength, weight, durability, and cost.

- Stainless steel is used for the frame and clamp.
- Nuts and bolts are used to join all parts.
- Rubber pads are used to improve grip on the pillar.
- Belt and tube are used for user support.

3.5 Fabrication of Mechanical Structure

The fabrication process involved the following steps:

Cutting and Shaping of Frame

- Stainless steel sheets and rods were cut according to the dimensions taken from CAD drawings.
- Bending operations were carried out to obtain the required frame shape.
- Sharp edges were removed using grinding to ensure safety during use.
- Holes were drilled at required locations for bolts, screw mechanism, and clamp assembly.

Frame Assembly

- All frame members were positioned and aligned properly before joining.

- Welding and nut-bolt connections were used to assemble the frame securely.
- Care was taken to maintain proper balance and uniform load distribution.
- The assembled frame was checked for rigidity and correct alignment.

Installation of Clamping Mechanism

- The adjustable clamp was fixed to the frame.
- The screw and nut were fitted for tightening action.
- Rubber pads were attached to improve grip.
- Clamp movement was checked manually.

Final Checking and Finishing

- All bolts and joints were tightened properly.
- The structure was checked for strength.
- Smooth working of the clamp was confirmed.
- The model was prepared for testing.

Chapter 4

Mechanical Design and Components

The adjustable pillar climber is a fully manual device designed to help users safely climb vertical pillars. It operates without motors, electronics, or external power sources, making the design simple and reliable. The climber uses friction-based grip pads to hold the pillar firmly during climbing. Its flexible frame allows adjustment for different square pillar sizes, making it suitable for various construction conditions. The lightweight and durable structure ensures easy handling, long service life, and stable operation with full user control.

4.1 Introduction to Mechanical Design

- Mechanical design is important for safe working of the pillar climber.
- The design focuses on strength, stability, and user safety.
- A fully manual mechanism is used without motors or power sources.
- The design uses simple parts for easy operation and fabrication.

4.2 Design Considerations

Before finalizing the mechanical layout, several key factors were studied:

Load-Bearing Capacity

- The pillar climber is designed to carry the weight of the user safely.
- The frame, clamp, and screw mechanism are designed to withstand climbing loads.
- Load is distributed evenly on the structure to avoid imbalance.
- This ensures safe operation during upward and downward movement.

Figure 4.1: CAD Model Of Frame

4.3 CAD Modeling and Structural Layout

The entire machine was first modeled in Solidedge to visualize dimensions, weight distri bution, and component placement.

- Key features observed in the CAD model:
- Arectangular lightweight base frame

CAD modeling helped verify:

- Proper fit of all components
- Correct alignment of parts
- Smooth working of the clamp
- Overall stability of the frame

4.4 Major Mechanical Components

The base frame acts as the main support structure for the adjustable pillar climber. It is fabricated using:

- Stainless steel or mild steel for sufficient strength

- Lightweight design to reduce overall load

Functions:

- Supports the adjustable clamp and screw mechanism
- Provides structural strength and rigidity
- Maintains proper alignment during climbing

4.5 Selection of Mechanical Materials

The materials were chosen based on strength,corrosion resistance, and cost.

4.6 Fabrication Process

The mechanical fabrication steps include:

4.7 Cutting and Shaping

- Aluminum or MS bars were cut to the required dimensions.
- Holes were drilled using a bench drilling machine.
- All edges were filed and finished for safe handling.

4.8 Frame Assembly

- All frame members were aligned properly.
- Parts were joined using welding and nut-bolt connections.
- Care was taken to maintain proper shape and balance.
- The assembled frame was checked for strength and alignment.

4.9 Working Principle of the Mechanical System

- The system works using a fully manual mechanism.
- No motors or electrical components are used.
- A screw is rotated by hand to operate the clamp.

- The clamp tightens around the pillar when the screw is turned.
- Grip pads create friction between the climber and the pillar.
- The grip prevents slipping during climbing.
- The user applies body weight gradually for movement.
- The climber is moved upward in small steps.
- The clamp is loosened slightly for repositioning.
- Controlled movement ensures safe climbing.

4.10 Advantages of the Mechanical Design

- Simple construction with fewer mechanical parts.
- Fully manual operation without electricity or fuel.
- Low cost and easy fabrication.
- Safe and stable during climbing.
- Easy to operate with minimum training.
- Lightweight and easy to carry.
- Requires low maintenance.
- Suitable for different pillar sizes.

4.11 Limitations of the Mechanical System

- Manual operation requires physical effort.
- Climbing speed is slow.
- Not suitable for very tall pillars.
- Depends on user strength and skill.

4.12 Summary

The adjustable pillar climber is a simple and fully manual mechanical system designed to improve safety during vertical climbing. The design avoids motors and electronics, which reduces complexity, cost, and chances of failure. An adjustable clamp with friction pads provides a firm grip on pillars of different sizes, ensuring stability during movement. The lightweight yet strong frame allows easy handling while supporting the user's weight. Overall, the system offers a practical, reliable, and economical solution for basic pillar climbing applications, with scope for further improvements in comfort and efficiency.

Chapter 5

System Hardware

The technical section is the most work-specific, and hence is the least described here.

Terminology: Define each term/symbol before you use it, or right after its first use. Stick to a common terminology throughout the report.

5.1 Body

The body of the adjustable pillar climber acts as the major support structure that holds all other components. The body of the structure has been designed using stainless steel, which has been chosen for this application due to its high strength, stiffness, and corrosion resistance properties. This structure keeps the mechanism in shape as well as ensures that all forces are equally distributed as a result of climbing. In addition, it also contains all other attachment points for the rest of the components, including the clamp, lead screw, nut body, as well as the handle. The design of the body ensures that it has a low center of gravity that provides stability to the climber.

Figure 5.1: Body

5.2 Bolt

The nut plays a critical role in the power transmission mechanism, where it translates the rotational motion of the lead screw into linear motion. In this project, the nut has been incorporated into the clamping system, which ensures effortless tightening and release of the jaws. With the commencement of screw rotation, it moves over the threads, hence developing a resultant force that retains the pillar. The choice of nut material, which aligns with that of the screw, takes into consideration factors of strength, durability, and compatibility.

Figure 5.2: Bolt

5.3 Clamp

The adjustable clamp is the functional unit of the climbing mechanism. The design of this mechanism can accommodate different diameters of pillars, as it adjusts through variations in the opening of the jaws. While turning the lead screw, this mechanism tightens around the pillar, and with theWOODEN OR RUBBER PADSThese pads are attached to the surface of the jaws, providing a grip to the pillar. The design of this mechanism ensures that it provides even pressure around the pillar, which prevents it from slipping. This mechanism is also able to accommodate different pillars due to its adjustable nature. This affects the functionality of climbing mechanism.

Figure 5.3: Clamp

5.4 Belt

The adjustable pillar climber features a high strength industrial belt, which is of major significance in grip and holding of the device on square pillars. This industrial belt usually consists of a reinforced nylon and rubber composite, which has high tensile strength. The design of this industrial belt has a texture that promotes grip, which ensures that it holds grip on the pillar when climbing. This assists in applying equal pressure on the pillar to avoid slippage.

Figure 5.4: Belt

Chapter 6

System Software

6.1 About Software Requirements

6.2 About Algorithm

6.3 About Flowchart

Chapter 7

Results

7.1 Fabrication Outcome

The adjustable pillar climber was successfully fabricated as per the planned design using a fully manual mechanical system. All major components such as the base frame, adjustable clamp, screw–nut mechanism, grip pads, and belt support were assembled properly. The fabricated structure was compact, lightweight, and mechanically sound. The overall finish of the model was satisfactory, with proper alignment of components and no visible defects in joints or connections.

7.2 Structural Performance

During inspection, the frame showed good rigidity and strength under normal loading conditions. The stainless steel structure was able to support the user's body weight without noticeable bending or deformation. Welded joints and nut-bolt connections remained firm during operation. The load distribution across the frame helped maintain balance, which contributed to stable climbing performance.

7.3 Gripping and Stability Performance

The adjustable clamp mechanism performed effectively during testing. The screw–nut arrangement allowed smooth tightening and loosening of the clamp around the pillar. Rubber grip pads generated sufficient friction to prevent slipping during climbing. The climber maintained proper contact with the pillar surface, ensuring stability during both upward and downward movement. No sudden loss of grip was observed under normal operating conditions.

7.4 Ease of Operation

The manual operation of the pillar climber was found to be simple and easy to understand. The screw mechanism required minimal effort to operate and provided good control over the gripping force. The absence of motors or electrical components made the system

straightforward and reliable. The climber could be operated without special training, making it user-friendly.

7.5 Safety Observations

Safety was maintained throughout the testing process. The climber remained stable during use, and the manual control provided confidence to the user. The belt support improved balance and reduced the risk of sudden movement. The mechanical design minimized the chances of unexpected failure, as there were no powered components involved.

Chapter 8

Conclusion

The fabrication of the adjustable pillar climber was successfully completed as per the objectives of the project. The main aim of developing a simple, safe, and manually operated climbing device was achieved using basic mechanical principles. The design avoids the use of motors, electronics, or external power sources, which reduces complexity, cost, and maintenance requirements. The screw-nut operated adjustable clamp provided effective gripping on the pillar, ensuring stability during climbing operations.

The fabricated structure demonstrated adequate strength and rigidity under normal working conditions. The use of durable materials contributed to the overall reliability and lifespan of the device. The friction-based grip pads enhanced safety by preventing slipping during upward and downward movement. Manual operation allowed the user to have full control over the climbing process, which increased confidence and stability.

8.1 Scope for future work

Although the prototype is functional, many enhancements can be made to increase efficiency, safety, and user convenience. Some recommended future developments include:

1. Improved Stability and Load Capacity

   - Frame strength and load capacity can be increased for taller pillars.

2. Partial Automation

   - Motor assistance and basic sensors can be added to reduce user effort.

3. Ergonomic Improvements

   - Better harness and padded supports can improve user comfort.

4. Safety Enhancements

   - Emergency locking and overload protection can be included.

5. Material Optimization

   - Lightweight materials can be used to reduce overall weight.

6. Performance Analysis

   - Detailed calculations and simulations can be carried out.

7. Field Testing

   - Long-term testing can be done under real working conditions.

8. Commercial Design Improvements

   - Design can be refined for mass production and better appearance.

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Appendices

Appendix A

Sustainable Development Goals addressed

#	**SDG**	**Level**
1	No Poverty	2
2	Zero Hunger	1
3	Good Health and Well-being	3
4	Quality education	2
5	Gender Quality	2
6	Clean water and Sanitation	1
7	Affordable and Clean Energy	3
8	Decent work and Economic Growth	3
9	Industry, Innovation and Infrastructure	3
10	Reduced Inequalities	2
11	Sustainable cities and Communities	3
12	Responsible Consumption and production	2
13	Climate action	2
14	Life below water	1
15	Life on Land	1
16	Peace, Justice and Strong Institutions	1
17	Partnership's for the Goals	2

Levels: Poor:1, Good :2, Excellent:3

Appendix B

Self-Assesment of the Project

Table 2.1: PO and PSO Contribution from the Project

#	**PO and PSO**	**Contribution from the Project**	**Level**
1	Engineering Knowledge	Application of mechanical fundamentals.	3
2	Problem Analysis	Analysis of safety and climbing issues.	3
3	Design/development of solutions	Design of adjustable climbing mechanism.	3
4	Conduct investigations of complex problems	Testing of grip and stability.	2
5	Modern tool usage	CAD modeling and drafting.	2
6	The Engineer and the world	Focus on safety and sustainability.	2
7	Ethics	Safe and responsible design.	2
8	Individual and Team Work	Team-based project execution.	3
9	Communication	Report and presentation preparation.	2
10	Project Management and Finance	Planning and cost control.	2
11	Life-long Learning	Learning through fabrication.	2
1	PSO1	Application of core engineering concepts.	3
2	PSO2	Practical system design and fabrication.	3

PSO1:Apply the concepts of electronic circuits and systems to analyses and design systems related to Microelectronics, Communication, Signal processing and Embedded sys-

tems for solving real world problems

PSO2:To identify problems in the area of communication and embedded systems and provide efficient solutions using modern tools/algorithms working in a team

Levels: Poor:1, Good :2, Excellent:3

Appendix C

Data Sheet of component 1

Table: General Details

Sl. No.	Parameter	Specification
1	Component Name	Adjustable Clamp
2	Material	Stainless Steel
3	Manufacturing Process	Cutting, Bending, Welding
4	Shape	Curved / Rectangular
5	Surface Finish	Smooth / Polished

Table:Dimensional Details

Sl. No.	Parameter	Specification
1	Length	As per design
2	Width	As per design
3	Thickness	2–4 mm
4	Weight	Approx. 1–1.5 kg
5	Mounting Type	Bolt and Nut

Appendix D

Data Sheet of component 2

Table: Functional and Performance Details

Sl. No.	**Parameter**	**Specification**
1	Function	Provides grip on pillar
2	Load Capacity	Supports user load
3	Adjustment Mechanism	Screw–nut system
4	Safety Factor	Adequate for manual use
5	Application	Manual pillar climbing